

The effects of Chinese herbal feed additives on physiological health and detoxification ability in the red claw crayfish, *Cherax quadricarinatus*, and evaluation of their safety

Xiaohui Shi^{a,1}, Yongyi Jia^{b,1}, Zhuofan Zhang^a, Wenbo Wu^a, Zijie Wu^a, Meili Chi^b, Qun Zhao^{a,*}, Erchao Li^{a,*}

^a Key Laboratory of Tropical Hydrobiology and Biotechnology of Hainan Province, Hainan Aquaculture Breeding Engineering Research Center, College of Marine Sciences, Hainan University, 58 Renmin road, Haikou, Hainan 570228, China

^b Key Laboratory of Healthy Freshwater Aquaculture, Ministry of Agriculture and Rural Affairs, Key Laboratory of Fish Health and Nutrition of Zhejiang Province, Zhejiang Institute of Freshwater Fisheries, Huzhou 313001, China

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ABSTRACT

The side effects of Chinese herbal medicine on metabolism in aquatic animals have not been fully studied. This study focused on evaluating the effects of *Astragalus* polysaccharide, chlorogenic acid, and berberine on the physiological health and detoxification ability of *Cherax quadricarinatus*. A basic diet served as a control, and diets containing differing concentrations of *Astragalus* polysaccharide (1, 2, 4, 8 g/kg), chlorogenic acid (1, 2, 4, 8 g/kg), or berberine (0.1, 0.5, 2.5 g/kg) were prepared. One thousand and eight crayfish (28.83 ± 0.29 g) were randomly allocated to 12 groups of net cages for 14 days. Samples were taken 7 days after feeding with Chinese herb bioactive ingredients and 7 days after ceasing this feeding. The results showed that compared with the control group, the addition of *Astragalus* polysaccharide, chlorogenic acid, or berberine did not have a significant effect on the growth performance of each experimental group. Cellular immunity, humoral immunity and antioxidant enzyme activities were increased significantly, while the concentrations of three herbal bioactive ingredients in plasma and metabolic enzymes improved significantly. Moreover, the bioactive ingredients did not cause biological macromolecular damage or histological damage in the hepatopancreas and significantly reduced the content of malondialdehyde. However, high concentrations of berberine (2.5 g/kg) significantly increased alanine aminotransferase and aspartate aminotransferase activities in the hepatopancreas, which could have a potential risk of damage to the hepatopancreas. Our results indicated that *Astragalus* polysaccharide, chlorogenic acid, and 0.1 g/kg berberine can be used as safe feed additives to stimulate the immune response, antioxidant capacity and detoxification ability and protect the organism from damage under oxidative stress.

1. Introduction

The red claw crayfish, *Cherax quadricarinatus*, is one of the world's most significant commercial freshwater species (Edgerton, 2005) and is expected to have a Chinese market value of approximately 9.3 billion USD by 2024 (Liu et al., 2020). However, as aquaculture operations

continue to increase in terms of both scale and stocking density, water quality is expected to deteriorate, and self-pollution is expected to intensify; such conditions are likely to lead to an increased risk of disease outbreaks in aquaculture operations (Shun et al., 2020; Safari and Paolucci, 2017). At present, antibiotics and synthetic drugs are still the main means to address diseases in most countries. However, the abuse of

Abbreviations: APS, *Astragalus* polysaccharides; CGA, chlorogenic acid; BBR, berberine; AOAC, association of official analytical chemists; WG, weight gain; SGR, specific growth rate; HSI, hepatosomatic index; CF, condition factor; PO, phenol oxidase; ProPO, prophenoloxidase; CAT, catalase; GPx, glutathione peroxidase; SOD, superoxide dismutase; ROS, reactive oxygen species; CYP450, cytochrome P450; GST, glutathione sulfo transferase; MDA, malondialdehyde; ALT, alanine aminotransferase; AST, aspartate aminotransferase; DsDNA, double-stranded DNA; SsDNA, single-stranded DNA; AuDNA, alkaline unwound DNA; PC, protein carbonyl; HLS, hemocyte lysate supernatant; ANOVA, analysis of variance.

* Corresponding authors.

E-mail addresses: zhaoq@hainanu.edu.cn (Q. Zhao), eceli@bio.ecnu.edu.cn (E. Li).

¹ These authors contributed equally to this manuscript.

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antibiotics not only affects the physiological health of aquatic animals but also may pose risks to the ecological environment. Moreover, antibiotic residues in aquatic products may also compromise food safety and cause potential harm to human health (Liu et al., 2017; Tong et al., 2020). Therefore, it is urgent to find a safe and effective alternative method to improve the immunity and health of aquatic animals in response to disease outbreaks.

Chinese herbs have been used to treat diseases in Asia for thousands of years (Wang and Ren, 2002). Their bioactive ingredients are considered effective alternatives to antibiotics, chemicals, vaccines, and other synthetic compounds. Hence, they have long been applied as attractants, growth promoters, antibacterial agents and immune preventive agents in aquaculture (Saad et al., 2006; Luo, 2012; Wang et al., 2010). Previous studies have shown that Chinese herbal bioactive ingredients, such as *Astragalus* polysaccharides (APS), chlorogenic acid (CGA), and berberine (BBR), could improve the growth performance and the immune response of aquatic animals. *Astragalus* polysaccharide is a polysaccharide component isolated from the root of *Astragalus membranaceus*, which is composed of glucose, galactose, etc. (Chang et al., 2018) and has been suggested to significantly promote the growth and activities of lysozyme and antioxidant enzymes in *Litopenaeus vannamei* and to enhance the specific growth rate (SGR), amylase and digestive enzyme activity of *Apostichopus japonicus* (Angela et al., 2020; Zhai et al., 2019; Huang et al., 2018; Sun et al., 2009). Chlorogenic acid is a phenolic compound widely distributed in the leaves of *Eucommia ulmoides*. It is a phenolic acid produced by caffeic acid and quinic acid. It has antioxidant and antibacterial effects and can significantly increase the feed conversion ratio, weight gain (WG), biomass gain and glutathione peroxidase (GPx) and catalase (CAT) activities in *L. vannamei* (Wang et al., 2013; Wang et al., 2015). Furthermore, berberine is the most abundant and representative compound in the alkaloids of *Coptis chinensis* Franch, and it has an antioxidative stress effect and has been applied to improve weight gain and the activities of lysozyme and peroxidase and to enhance the component 3 of *Cyprinus carpio* and *Megalobrama amblycephala* (Chen et al., 2016; Zhou et al., 2016; Xu et al., 2017). Thus, it is believed that Chinese herb bioactive ingredients have wide application prospects in aquaculture as a substitute for antibiotic drugs. However, with the continuous expansion of the species and the range of Chinese herb bioactive ingredients used in aquaculture in recent years, the possibility of adverse biological reactions has also gradually expanded (Pu et al., 2017). It is extremely necessary to study the damaging effects and detoxification of Chinese herb bioactive ingredients in aquatic animals. Comprehensive studies on the damaging effects and detoxification of Chinese herbal medicine additives in aquatic animals are urgently needed.

Various studies have shown that the metabolism of exogenous substances in organisms is divided mainly into two stages. Phase I metabolism mainly catalyzes the monooxygenation of endogenous and exogenous compounds through cytochrome P450 (CYP450) (Hu et al., 2015). Then, phase II metabolism mainly catalyzes the combination of active groups with exogenous substances and active metabolites through metabolic enzymes such as glutathione sulfo transferase (GST) to form water-soluble metabolites to discharge from the organism (Keen and Jakoby, 1978; Hayes and Wolf, 1988). Drug metabolism has a certain time cycle, and metabolites with short half-lives can easily cause idiosyncratic adverse reactions (Okano et al., 1979). However, there are few studies on the accumulation, metabolic detoxification and damaging effects of Chinese herb bioactive ingredients in aquatic animals; most of the research focuses on mammals. To our knowledge, few studies have investigated related metabolic processes and biological damage in red claw crayfish, and only one study identified the effects of *Astragalus membranaceus* and *Coptis chinensis* on biomolecule damage in *L. vannamei* (Ding et al., 2020). The lack of research on the metabolic process and safety evaluation of Chinese herb bioactive ingredients in aquatic animals has greatly limited the wide application of Chinese herbal medicine in *C. quadricarinatus* culture.

Based on this background, this study investigated the effects of Chinese herbal feed additives on the growth performance, immune function, antioxidant capacity, and detoxification in *C. quadricarinatus* by setting different concentration gradients of *Astragalus* polysaccharides, chlorogenic acid and berberine and comprehensively evaluated the safety of the Chinese herb bioactive ingredients to determine the appropriate doses. This study will provide a technical reference for the application of Chinese herbal medicine as an effective and safe immunopotentiator in aquatic animals and promote the healthy development of *C. quadricarinatus* culture.

2. Materials and methods

2.1. Experimental diets

Astragalus polysaccharides and berberine were obtained from Ruidi Biotechnology Co., Ltd. (Xi'an, China), with purities of 90% and 97%, respectively. Chlorogenic acid was obtained from Xingyun Biological Technology Co., Ltd. (Hangzhou, China), with a purity of 98%.

Diet formulations are presented in Table 1. *Astragalus* polysaccharides and chlorogenic acid were supplemented separately into the basic diet at the expense of cellulose to obtain final doses of *Astragalus* polysaccharides and chlorogenic acid of 1, 2, 4 and 8 g/kg, respectively. Berberine was supplemented separately into the basal diet at the expense of cellulose to obtain final berberine doses of 0.1, 0.5 and 2.5 g/kg. Butylated hydroxytoluene at a concentration of 0.05‰ was added as an antioxidant to avoid lipid oxidation in diets.

All ingredients were ground with a 60-μm mesh and then mixed. The oil was added to the dry mixture, and the appropriate amount of distilled water was added and mixed well in a mesh. Pelleting was carried out through a double-helix plodder (CD4 × 1TS, SCUT industrial factory, Guangdong, China). The pellets were broken up to a proper size (2.5 mm pellet diameter) through mincer attachment. The pellets were then dried in a well-ventilated room for approximately 24 h and stored in a freezer at −20 °C until use.

2.2. Experimental design

Crayfish were obtained from the Guangxi Nanfan Agricultural Breeding Base. Crayfish were fed a basic diet twice per day and adapted to the experimental conditions for one week. After acclimation, the crayfish (initial weight 28.83 ± 0.29 g) were randomly distributed in 36 net cages (1.0 m length × 0.5 m width × 1 m height) with three replicates of one treatment at a density of 30. Net cages were placed in a cement pool provided with continuous aeration. The 14-day experiment was divided into two stages: Stage 1 was feeding with Chinese herb bioactive ingredients for 7 days, and stage 2 was the 7 days after this feeding was ceased. During the experimental period, the temperature of the water, pH, dissolved oxygen, salinity, total ammonia nitrogen and nitrite were monitored daily and recorded as 24–27 °C, 7.8–8.2, 4 mg/L, 3‰, 0.25 mg/L–0.28 mg/L and 0.18–0.2 mg/L, respectively. Feces and uneaten feed were regularly removed from the net cage, and approximately 60% of the water in the cement pool was replaced every 3 days to maintain a normal level.

2.3. Sample collection

Twelve crayfish from each test group were sampled on Day 7 of feeding with Chinese herb bioactive ingredients and on Day 7 after this feeding was ceased, and the crayfish were starved for 24 h before sampling.

Briefly, the hemolymph (0.25 mL) of each crayfish was drawn from the central sinus with a sterilized syringe containing the anticoagulant (NaCl 0.14 M, C₆H₁₂O₆·2H₂O 0.1 M, C₆H₅Na₃O₇·2H₂O 0.03 M, C₆H₁₀O₈ 0.026 M, EDTA-2Na 0.01 M, pH = 4.6) (Lin, 2017) of 0.25 mL and then centrifuged at 800 ×g for 10 min, and the resulting supernatant (plasma)

Table 1

Formulation of the experimental diets (g/ kg dry matter basis).

Ingredient	control	Astragalus polysaccharide (APS)				Chlorogenic acid (CGA)				Berberine (BBR)		
		1	2	4	8	1	2	4	8	0.1	0.5	2.5
Fish meal	300	300	300	300	300	300	300	300	300	300	300	300
Soybean meal	300	300	300	300	300	300	300	300	300	300	300	300
Wheat starch	200	200	200	200	200	200	200	200	200	200	200	200
Soybean oil	50	50	50	50	50	50	50	50	50	50	50	50
Choline chloride	10	10	10	10	10	10	10	10	10	10	10	10
Sodium carboxymethyl cellulose	25	25	25	25	25	25	25	25	25	25	25	25
Cholesterol	5	5	5	5	5	5	5	5	5	5	5	5
Lecithin	10	10	10	10	10	10	10	10	10	10	10	10
Herbal extracts	0	1	2	4	8	1	2	4	8	0.1	0.5	2.5
Cellulose	60	59	58	56	52	59	58	56	52	59.9	59.5	57.5
Butylated hydroxytoluene	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
Vitamin premix ^a	20	20	20	20	20	20	20	20	20	20	20	20
Mineral premix ^b	20	20	20	20	20	20	20	20	20	20	20	20
Total	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000
Proximate composition (g/ kg)												
Moisture	92.3	93.8	90.4	89.8	92.6	87.1	88.9	90.3	86.9	85.5	88.4	89.5
Crude protein	360.4	360.0	361.2	361.0	360.9	359.7	359.8	360.7	360.4	360.2	361.4	360.4
Crude lipid	111.3	112.5	109.8	110.6	113.2	114.2	112.7	11.1	109.8	113.2	110.6	111.9
Crude ash	80.2	79.3	78.5	79.9	76.8	79.1	80.5	76.4	77.9	79.2	76.4	78.5

^a Vitamin premix (per 100 g premix): thiamin, 0.05 g; riboflavin, 0.3 g; pyridoxine, 0.1 g; cyanocobalamin, 0.03 g; ascorbic acid (35%), 10 g; alpha-tocopherol (50%), 5 g; menadione, 0.2 g; inositol, 0.5 g; nicotinamide, 0.5 g; cholecalciferol (500,000 IU/g), 0.08 g; retinol palmitate (500,000 IU/g), 0.05 g; folic acid, 0.02 g; biotin, 0.005 g; choline chloride, 10 g; pantothenic acid, 0.5 g. All ingredients were filled with α -cellulose to 100 g.

^b Mineral premix (per 100 g premix): KCl, 2.8 g; MgSO₄, 7.1 g; ZnSO₄, 2 g; MnSO₄·H₂O, 0.162 g; CuSO₄·5H₂O, 0.06 g; Ca(IO₃)₂, 0.02 g; CoCl₂·6H₂O, 0.01 g; NaH₂PO₄, 21.5 g; CaHPO₄·2H₂O, 25 g; CaCO₃, 18 g; Na₂SeO₃, 0.025 g; FeSO₄·H₂O, 1.0 g; and KH₂PO₄, 21.6 g. All ingredients were diluted with α -cellulose to 100 g.

and hemocytes were collected and stored at -80°C . Plasma was used to measure phenol oxidase activity, bacteriolytic activity, antibacterial activity and agglutination activity, and hemocytes were used to measure hemocyte respiratory burst. Preparation of hemocyte lysate supernatant (HLS) was based on the method proposed by Smith and Söderhäll (1991). The resulting hemocyte precipitates were rinsed and resuspended in sterilized crayfish normal saline (NaCl 0.2 M, KCl 5.4 mM, CaCl₂·2H₂O 10 mM, MgCl₂·6H₂O 2.6 mM and NaHCO₃ 2 mM, pH = 6.8) (Lin, 2017). After centrifugation, the hemocytes were resuspended again in 0.6 mL sterilized saline and then crushed ultrasonically at 20 W $\times$ 30% for 60 s. The resultant fluid (HLS) was then stored at -80°C for the measurements of the prophenoloxidase activity of hemocytes.

The hepatopancreas was separated and quickly preserved in liquid nitrogen and then stored at -80°C for further analysis. The hepatopancreases were collected and homogenized (1:9, w/v) in 0.9% cold NaCl. The homogenates were then centrifuged (4 $^{\circ}\text{C}$, 10 min) with centrifugal force as required by the instructions. The supernatants were individually used to detect antioxidant capacity (superoxide dismutase, catalase and glutathione peroxidase), cytochrome P450, glutathione sulfo transferase, malondialdehyde, alanine aminotransferase and aspartate aminotransferase. The protein contents of the hepatopancreas

to constant weights at 105 $^{\circ}\text{C}$ for moisture determination. Crude lipid contents were determined with a Soxhlet system, and the ash content was determined by heating in a muffle (SX2-4-10 N, Yiheng) furnace at 550 $^{\circ}\text{C}$ for 6 h.

2.5. Growth performance parameters assay

The crayfish in each cage were counted to calculate survival and then weighed to evaluate weight gain, specific growth rate and hepatosomatic index (HSI). Body length was measured to evaluate the condition factor (CF). The survival, weight gain, specific growth rate, hepatosomatic index and condition factor were calculated with the following equations (Chen et al., 2021):

$$\text{Survival (\%)} = (\text{final number of crayfish} / \text{initial number of crayfish}) \times 100.$$

$$\text{Weight gain (\%)} = (\text{final weight} - \text{initial weight}) / \text{initial weight} \times 100.$$

$$\text{Specific growth rate (\%day}^{-1}\text{)} = [\ln(\text{final weight}) - \ln(\text{initial weight})] \times 100 / \text{days}.$$

$$\text{Hepatosomatic index (\%)} = (\text{wet hepatopancreatic weight} / \text{wet body weight}) \times 100.$$

were quantified using the Bradford method, described in 1976 (Bradford, 1976).

2.4. Whole-body composition analysis

The dietary composition was determined according to the standard methods of the AOAC procedures (AOAC, 2005). Dumas combustion methods with a protein analyzer (Elementar rapid N exceed, Germany) were used to determine the crude protein contents. Samples were dried

$$\text{Condition factor (\%)} = \text{final weight} / \text{length}^3 \times 100.$$

2.6. Immune response parameters assay

Hemocyte respiratory bursts were detected according to the method provided by Song and Hsieh (1994); that is, the activity of respiratory bursts was calculated by using the reduction of formazan by

nitroterrazolium chloride (NBT) as an indicator for the production of superoxide anions.

The prophenoloxidase (proPO) activity of plasma was measured using L-3,4-dihydroxyphenylalanine as the substrate and trypsin (Sigma–Aldrich, USA) as the activator following the method described by Smith and Söderhäll (1991). Briefly, 100 µL of hemocyte lysate supernatant was incubated with 100 µL of 0.1% trypsin and pipetted into a 96-well ELISA plate, which was incubated in the dark at 25 °C for 30 min. Then, 100 µL of L-3,4-dihydroxyphenylalanine was added. Subsequently, the plate was incubated in the dark at 25 °C for 10 min. The absorbance was measured at 490 nm. One unit of enzyme activity was defined as an increase in absorbance of 0.001/min/mL hemocyte lysate supernatant.

Plasma phenol oxidase (PO) activity was measured using L-3,4-dihydroxyphenylalanine as a substrate following the method described by Mason (1956) and Hernández-López et al. (1996). Briefly, 150 µL of 2 mg/mL L-3,4-dihydroxyphenylalanine and 50 µL of plasma were pipetted into a 96-well ELISA plate. The absorbance was measured at 490 nm at an interval of 2 min for 30 min. One unit of enzyme activity was defined as an increase in absorbance of 0.001/min/mL.

Bacteriolytic activity in plasma was improved by Hultmark et al. (1980). Freeze-dried powder of *Micrococcus lysodeikticus* was used as the substrate to prepare a bacterial suspension with sterilized 0.1 M phosphate buffer solution. A total of 250 µL bacterial suspension and 50 µL plasma were pipetted into a 96-well ELISA plate. The absorbance (A_0) at 570 nm was immediately measured after vortexing. Subsequently, the plate was incubated in the dark at 37 °C and mixed well every 10 min for 30 min, and the optical density at 570 nm (A) was again measured. The bacteriolytic (U_L) activities were calculated from the following equations:

$$U_L = (A_0 - A)/A.$$

The antibacterial activity in plasma was modified by the method of Hultmark et al. (1980). The cryopreservation liquid *Vibrio parahaemolyticus* was thawed, inoculated, and then spread on solid 2116E. The bacteria were incubated at 28 °C for 20 h and were washed gently from the plate into a sterile tube with sterile 0.1 M phosphate buffer solution and then prepared into a suspension with a certain concentration ($OD_{570nm} = 0.3–0.5$). Briefly, 250 µL of bacterial suspension and 50 µL of plasma were pipetted into a 96-well ELISA plate, and the OD_{570nm} was read by a microplate reader and recorded as A_0 . Subsequently, the plate was incubated in the dark at 37 °C for 30 min, and then the OD_{570nm} was recorded as A . The antibacterial (U_A^2) activities were calculated from the following equations:

$$U_A^2 = (A_0 - A)/A$$

The method for determining plasma agglutination activity was modified by Pais et al. (2007). First, 10 mg/mL sodium heparin anticoagulant was prepared. Thirty milligrams of sodium heparin was weighed and placed in a 5 mL centrifuge tube, and 3 mL of ultrapure water was added to dissolve it. The sodium heparin was divided into ten 5 mL centrifuge tubes with 0.3 mL each. The mice were purchased from the Hainan Institute of Materia Medicine. Blood was drawn from the eyeballs of mice and added to centrifugal tubes containing 0.3 mL of sodium heparin anticoagulant, with approximately 3 mL of blood added to each tube. The plasma from each anticoagulant blood sample was removed by centrifugation at 2000 r/min for 10 min and fully washed with 0.9% NaCl normal saline (1:1). The supernatant was then removed by centrifugation at 2000 r/min for 3 min, which was repeated 4 times. Finally, the supernatant was removed by centrifugation at 2500 r/min for 3 min. The remaining erythrocytes and normal saline were prepared into a 2% erythrocyte suspension in a 1:49 ratio and stored at 4 °C for later use. The hemagglutinating activity was measured by incubating the erythrocytes with a 96-well plate. A total of 25 µL of TBS— Ca^{2+} was added to each well separately and then 25 µL of plasma to the first well. After mixing, 25 µL of the mixture was absorbed from the first well and added to the second well. Then, 25 µL of mice erythrocyte suspension

was added to each well and incubated at room temperature for 1–2 h to observe the erythrocyte agglutination status. The agglutination activity of plasma was detected. The results were observed 10 min after standing and could be observed after the erythrocytes in the control well had precipitated. All erythrocytes were agglutinated and deposited at the bottom of the pore. The erythrocytes were laid out in a network and could be 100% agglutinated. The nonagglutinated erythrocytes were deposited at the bottom of the pore in a punctate shape. The titer of plasma agglutination was 2^{n+1} .

2.7. Antioxidant capacity parameters assay

Superoxide dismutase (SOD), catalase and glutathione peroxidase were used in commercial kits (Nanjing Jiancheng Bioengineering Institute, Nanjing, China; A001–3-2, A007–1-1, A005–1-2). The kits used the WST-1 method, visible light method and colorimetric method. This test was carried out in accordance with the manufacturer's instructions.

2.8. Detoxification parameters assay

Astragalus polysaccharides standard curve drawing: First, 0.6, 0.9, 1.2, 1.5, 1.8 and 2.1 mL of standard glucose solution (1 g/L) was added to a 50 mL volumetric flask, ultrapure water was added to the scale, and the solution was well mixed. Second, 2 mL of the above solutions was added to 1 mL of 5% phenol solution and 5 mL of concentrated sulfuric acid, shaken gently for 5 min, mixed evenly and left at room temperature for 30 min. Finally, the absorbance value was measured at 490 nm with ultrapure water as a blank control. The regression curve was made with the glucose concentration as the abscissa and the absorbance value as the ordinate.

The *Astragalus* polysaccharides content was determined by phenolsulfuric acid colorimetry. Two milliliters of plasma was added to 1 mL of 5% phenol solution and 5 mL of concentrated sulfuric acid, gently shaken for 5 min, mixed evenly, placed at room temperature for 5 min, boiled in a water bath for 15 min, cooled to room temperature with running water, and ultrapure water was used as a blank control, and the absorbance value was measured at 490 nm (Cao et al., 2009).

Pretreatment of the chlorogenic acid and berberine samples: A total of 150 µL acetonitrile was added to 150 µL plasma and centrifuged at 5000 r/min for 15 min. The supernatant was transferred through a 0.22 µm organic filter membrane to a nitrogen blower on a 1.5 mL sample bottle for nitrogen blowing, and 300 µL methanol was added.

Determination of chlorogenic acid content was performed according to existing methods (Qi et al., 2011). The chlorogenic acid content was determined by high-performance liquid chromatography (Agilent 1100) with a C18 column (4.6 × 250 mm). The mobile phase consisted of an aqueous solution of acetonitrile–0.4% phosphoric acid (1:9), an injection volume of 50 µL, a detection wavelength of 270 nm, a flow rate of 0.8 mL/min and a column temperature of 35 °C.

Determination of berberine content was performed according to existing methods (Friedemann et al., 2014) with modifications. The berberine content was determined by high-performance liquid chromatography (Agilent 1100) with a C18 column (4.6 × 250 mm). The mobile phase consisted of water with acetonitrile–0.4% phosphoric acid (25:75), an injection volume of 50 µL, detection wavelength of 265 nm, flow rate of 1 mL/min and column temperature of 35 °C.

Cytochrome P450 was measured with an ELISA kit (Shanghai Fantasy Biological Technology Co., Shanghai, China; FT-TX21675P). Glutathione sulfo transferase was measured with a commercial kit (Nanjing Jiancheng Bioengineering Institute, Nanjing, China; A004–1-1), and the kit used the colorimetric method. The measurements were carried out according to the manufacturers' instructions.

2.9. Damage parameters assay

2.9.1. Molecular damage

The rate of transition of double-stranded DNA (dsDNA) to single-stranded DNA (ssDNA) under predefined alkaline denaturing conditions was proportional to the number of breaks in the phosphodiester backbone and hence was used as a measure of DNA integrity (Daniel et al., 1985). The assay was adapted from previously described methods of DNA extraction and alkaline unwinding (Miao et al., 2009; Ching et al., 2001). The DNA sample was diluted and divided into three equal portions for fluorescence determination of alkaline unwound DNA (auDNA), dsDNA, and ssDNA. The ratio between double-stranded DNA and total DNA (F value) was determined as follows:

$$F \text{ value} = (\text{auDNA} - \text{ssDNA})/(\text{dsDNA} - \text{ssDNA}).$$

Protein carbonyl (PC) content was evaluated by measuring the formation of protein hydrazones resulting from the dinitrophenyl hydrazine reaction with protein carbonyls. The carbonyl content was measured at 375 nm. In addition, the total protein content of hepatopancreas samples was measured in diluted homogenates by the Bradford method using bovine serum albumin (Sigma) as a standard.

Commercial kits were used to detect malondialdehyde (MDA), alanine aminotransferase (ALT) and aspartate aminotransferase (AST) (Nanjing Jiancheng Bioengineering Institute, Nanjing, China; A003-1-2, C009-2-1, C010-2-1). The kits used the TBA method, Reitman-Frankel method and Reitman-Franke method. The measurements were carried out in accordance with the manufacturer's instructions.

2.9.2. Hepatopancreas histology

The hepatopancreases were collected from crayfish randomly selected from the treatment groups and fixed with 4% paraformaldehyde for 24 h. The hepatopancreases were washed with water and dehydrated with different concentrations of ethanol. Then, the hepatopancreases were placed into xylene for transparency and embedded in paraffin. Wax blocks were sectioned at 5 μm using a rotary microtome (Leica RM2105, Germany). After dewaxing, the slices were stained with hematoxylin and eosin. The sections were observed and photographed by microscopy (ECLIPSE 200, Nikon, Japan).

2.10. Quantitative real-time PCR

Total RNA of hemocytes and hepatopancreas of the control and treatment groups were extracted by TRIzol Reagent (Invitrogen, USA). The primer sequences for qRT-PCR are shown in Table 2, and β -actin was used as the housekeeping gene. The total volume of the amplification reaction system was 20 μL , and the reaction mixture was composed of 10 μL SYBR Master mix (2 $\times$) and 0.4 μL (each) forward and reverse primers with a concentration of 10 μM , 7.2 μL dd- H_2O , and 2 μL cDNA. Quantitative PCR was performed at 95 $^\circ\text{C}$ for 5 min, followed by 40 cycles at 95 $^\circ\text{C}$ for 15 s and 60 $^\circ\text{C}$ for 30 s. At the end of each PCR, the melting curve of the amplification product was generated to ensure the specificity of the detection. The $2^{-\Delta\Delta\text{Ct}}$ method was used to analyze the relative expression (fold changes) of the target gene (Livak and Schmittgen, 2001).

Table 2
Primer information.

Gene	Position	qPCR primer pairs (5'-3')	GenBank accession number
CYP1A1	Forward	GCAGTTTGCACAGACGTGTT	DQ286837.1
	Reverse	CCATGCTGGGAAACGATTA	
β -actin	Forward	TTAGAAACACTTCCTGTGGACAATA	MN918115.1
	Reverse	GTATTTCAGTATGGATTGGAGTTCT	

2.11. Statistical analysis

One-way analysis of variance (ANOVA) and Duncan's multiple interval test were used to analyze each group's growth performance and biochemical indices, and the significant differences between the groups were compared. A t -test was used to analyze changes in the same treatment group at different time points. The above statistical analyses were conducted using SPSS software (ver. 26.0; SPSS Inc., USA). Data are expressed as the mean $\pm$ standard deviation (SD). $P < 0.05$ was regarded as statistically significant.

3. Results

3.1. Growth performance

Table 3 showed the effects of *Astragalus* polysaccharides, chlorogenic acid, and berberine on the growth performance of crayfish. There were no significant differences in survival, weight gain, specific growth rate, hepatosomatic index or condition factor between the experimental groups and the control group ($P > 0.05$), but the weight gain of all treatment groups was higher than that of the control group after 7 days of feeding and 7 days of this feeding being ceased.

3.2. Immune functions

The effects of *Astragalus* polysaccharides, chlorogenic acid, and berberine on the respiratory burst of hemocyte crayfish were shown in Fig. 1. Compared with the control group, feeding Chinese herbal medicines significantly increased respiratory burst ($P < 0.05$), among which the *Astragalus* polysaccharides 4 g/kg, chlorogenic acid 4 g/kg and berberine 0.1 g/kg groups were significantly enhanced. After 7 days of this feeding being ceased, respiratory burst was significantly reduced in the experimental groups but still significantly higher than that in the control group ($P < 0.05$).

The effects of *Astragalus* polysaccharides, chlorogenic acid, and berberine on the prophenoloxidase activation system of crayfish were shown in Fig. 2. Adding *Astragalus* polysaccharides, chlorogenic acid or berberine for 7 days significantly increased the activities of prophenoloxidase and phenol oxidase ($P < 0.05$). The effects of 2 g/kg *Astragalus* polysaccharides, 2 and 4 g/kg chlorogenic acid, and 0.1 g/kg berberine on the prophenoloxidase activation system in crayfish were significant. After 7 days of this feeding being ceased, prophenoloxidase activity and phenol oxidase activity in all experimental groups decreased significantly, while prophenoloxidase activity was still significantly higher than that in the control group ($P < 0.05$).

The results of bacteriolytic activity, antibacterial activity and agglutination activity of crayfish were presented in Fig. 3. The addition of *Astragalus* polysaccharides, chlorogenic acid, or berberine for 7 days significantly enhanced the bacteriolytic activity, antibacterial activity and agglutination activity ($P < 0.05$). The activities of the *Astragalus* polysaccharides 4 g/kg, chlorogenic acid 2, 4 g/kg, and berberine 0.1 g/kg groups were significant. After 7 days of this feeding being ceased, bacteriolytic activity, antibacterial activity and agglutination activity in all experimental groups decreased significantly, while the antibacterial activity was still significantly higher than that in the control group ($P < 0.05$).

3.3. Antioxidant capacity

Figure 4 clarified the effects of *Astragalus* polysaccharides, chlorogenic acid, and berberine on the antioxidant enzyme activities of crayfish. After 7 days of feeding, superoxide dismutase activity, catalase activity and glutathione peroxidase activity in the experimental groups were increased significantly compared with those in the control group ($P < 0.05$), among which the *Astragalus* polysaccharides 2, 4 g/kg, chlorogenic acid 2, 4 g/kg, and berberine 0.1 g/kg groups were

Table 3

Effects of *Astragalus* polysaccharides, chlorogenic acid, and berberine on the growth performance of *C. quadricarinatus* after feeding for 7 days and 7 days after feeding was ceased.

Treatment (g/ kg)	CON	Astragalus polysaccharide (APS)				P1	Chlorogenic acid (CGA)				P2	Berberine (BBR)			P3
		1	2	4	8		1	2	4	8		0.1	0.5	2.5	
Feeding for 7 days															
Survival (%)	100	100	100	100	100	1.00	100	100	100	100	1.00	100	100	100	1.00
Weight gain (%)	1.60	1.59	1.57	1.56	1.60	0.88	1.52	1.53	1.68	1.60	0.52	1.66	1.58	1.65	0.77
	± 0.41	± 0.26	± 0.36	± 0.23	± 0.39		± 0.16	± 0.23	± 0.30	± 0.17		± 0.12	± 0.35	± 0.08	
Specific growth rate	0.23	0.22	0.22	0.22	0.23	0.88	0.22	0.22	0.24	0.23	0.52	0.23	0.22	0.23	0.77
(%day ⁻¹)	± 0.06	± 0.04	± 0.05	± 0.03	± 0.05		± 0.02	± 0.03	± 0.04	± 0.02		± 0.02	± 0.05	± 0.01	
Hepatosomatic	5.44	5.49	5.57	5.68	5.69	0.45	5.79	5.72	5.77	5.68	0.23	5.64	5.57	5.60	0.37
index (%)	± 0.32	± 0.48	± 0.13	± 0.47	± 0.28		± 0.30	± 0.43	± 0.31	± 0.10		± 0.26	± 0.07	± 0.23	
Condition factor	2.49	2.57	2.58	2.49	2.60	0.60	2.54	2.53	2.67	2.56	0.43	2.65	2.60	2.56	0.56
(%)	± 0.28	± 0.22	± 0.28	± 0.26	± 0.16		± 0.23	± 0.15	± 0.28	± 0.29		± 0.38	± 0.16	± 0.38	
7 days after feeding ceased															
Survival (%)	100	100	100	100	100	1.00	100	100	100	100	1.00	100	100	100	1.00
Weight gain (%)	2.03	1.81	1.97	1.98	1.94	0.49	2.03	1.94	1.92	1.96	0.61	1.94	1.90	1.82	0.33
	± 0.23	± 0.47	± 0.45	± 0.18	± 0.29		± 0.32	± 0.24	± 0.21	± 0.25		± 0.16	± 0.34	± 0.16	
Specificgrowth rate	0.29	0.26	0.28	0.28	0.27	0.49	0.29	0.27	0.27	0.28	0.61	0.27	0.27	0.26	0.33
(%day ⁻¹)	± 0.03	± 0.07	± 0.06	± 0.03	± 0.04		± 0.05	± 0.03	± 0.03	± 0.04		± 0.02	± 0.05	± 0.02	
Hepatosomatic	5.72	5.72	5.69	5.76	5.61	0.64	5.64	5.71	5.70	5.64	0.78	5.59	5.72	5.74	0.61
index (%)	± 0.31	± 0.43	± 0.24	± 0.37	± 0.31		± 0.31	± 0.30	± 0.23	± 0.38		± 0.18	± 0.43	± 0.29	
Condition factor	2.61	2.51	2.57	2.55	2.62	0.67	2.58	2.56	2.52	2.57	0.66	2.57	2.48	2.52	0.59
(%)	± 0.16	± 0.28	± 0.25	± 0.25	± 0.42		± 0.17	± 0.20	± 0.21	± 0.30		± 0.36	± 0.17	± 0.34	

P1, ANOVA *P* value for the *Astragalus* polysaccharide group and control group; P2, ANOVA *P* value for the chlorogenic acid group and control group; P3, ANOVA *P* value for the berberine group and control group; APS, *Astragalus* polysaccharide; CGA, chlorogenic acid; BBR, berberine. Data are the mean ± SD.

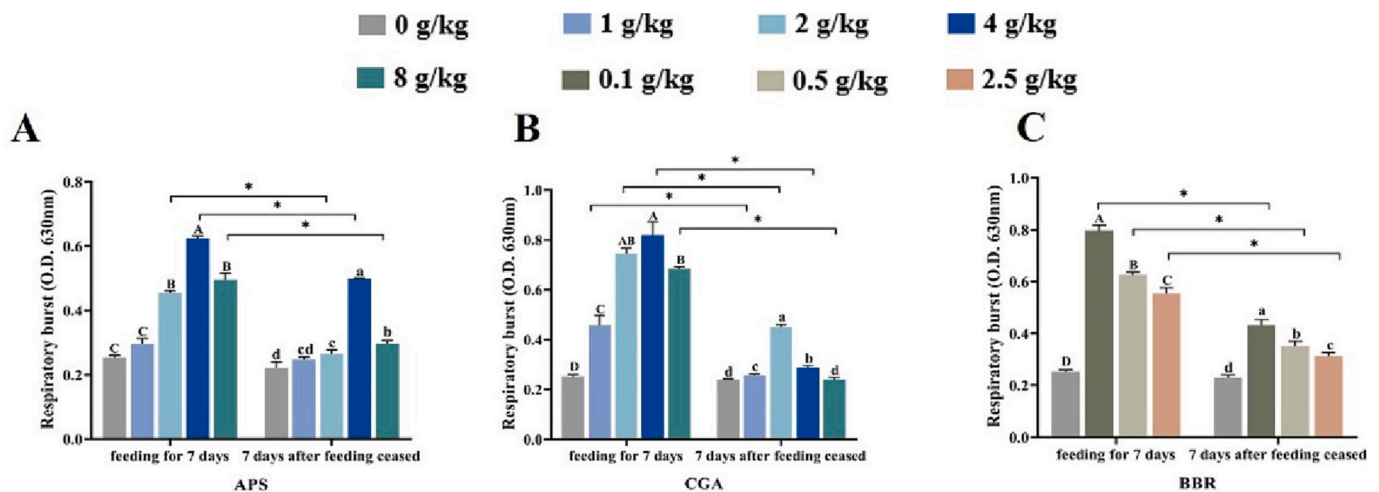


Fig. 1. Effects of *Astragalus* polysaccharides, chlorogenic acid and berberine on hemocyte respiratory burst of *C. quadricarinatus* after feeding for 7 days and 7 days after feeding was ceased. (A: APS hemocyte respiratory burst; B: CGA hemocyte respiratory burst; C: BBR hemocyte respiratory burst). APS, *Astragalus* polysaccharide; CGA, chlorogenic acid; BBR, berberine. Data are the mean ± SD. Bars with different letters represent significant differences ($P < 0.05$). Statistical significance is indicated by * ($P < 0.05$).

significantly enhanced. After 7 days of this feeding being ceased, the activities of antioxidant enzymes (superoxide dismutase, catalase, glutathione peroxidase) in the *Astragalus* polysaccharides 2, 4, 8 g/kg, chlorogenic acid 4, 8 g/kg, and berberine 0.1, 0.5, 2.5 g/kg groups were significantly higher than those in the control group ($P < 0.05$).

3.4. Detoxification

The effects of *Astragalus* polysaccharides, chlorogenic acid, and berberine on the plasma concentration of crayfish were shown in Fig. 5. After 7 days of feeding, compared with the control group, the plasma drug concentrations in all experimental groups were significantly enhanced ($P < 0.05$). After 7 days of this feeding being ceased, the plasma drug concentration in all treatment groups decreased significantly but was still significantly higher than that in the control group (P

< 0.05).

The effects of crayfish metabolic enzyme reactions were presented in Fig. 6. After 7 days of feeding, compared with the control group, the cytochrome P450 concentration, expression of CYP1A1 mRNA and glutathione sulfo transferase activity in all experimental groups were significantly enhanced ($P < 0.05$), among which the *Astragalus* polysaccharides 2, 4 g/kg, chlorogenic acid 2, 4 g/kg, and berberine 0.1 g/kg groups were significantly enhanced. After 7 days of this feeding being ceased, the metabolic enzyme activity and expression of CYP1A1 mRNA in all treatment groups decreased significantly but were still significantly higher than those in the control group ($P < 0.05$), except for the 1 g/kg chlorogenic acid group.

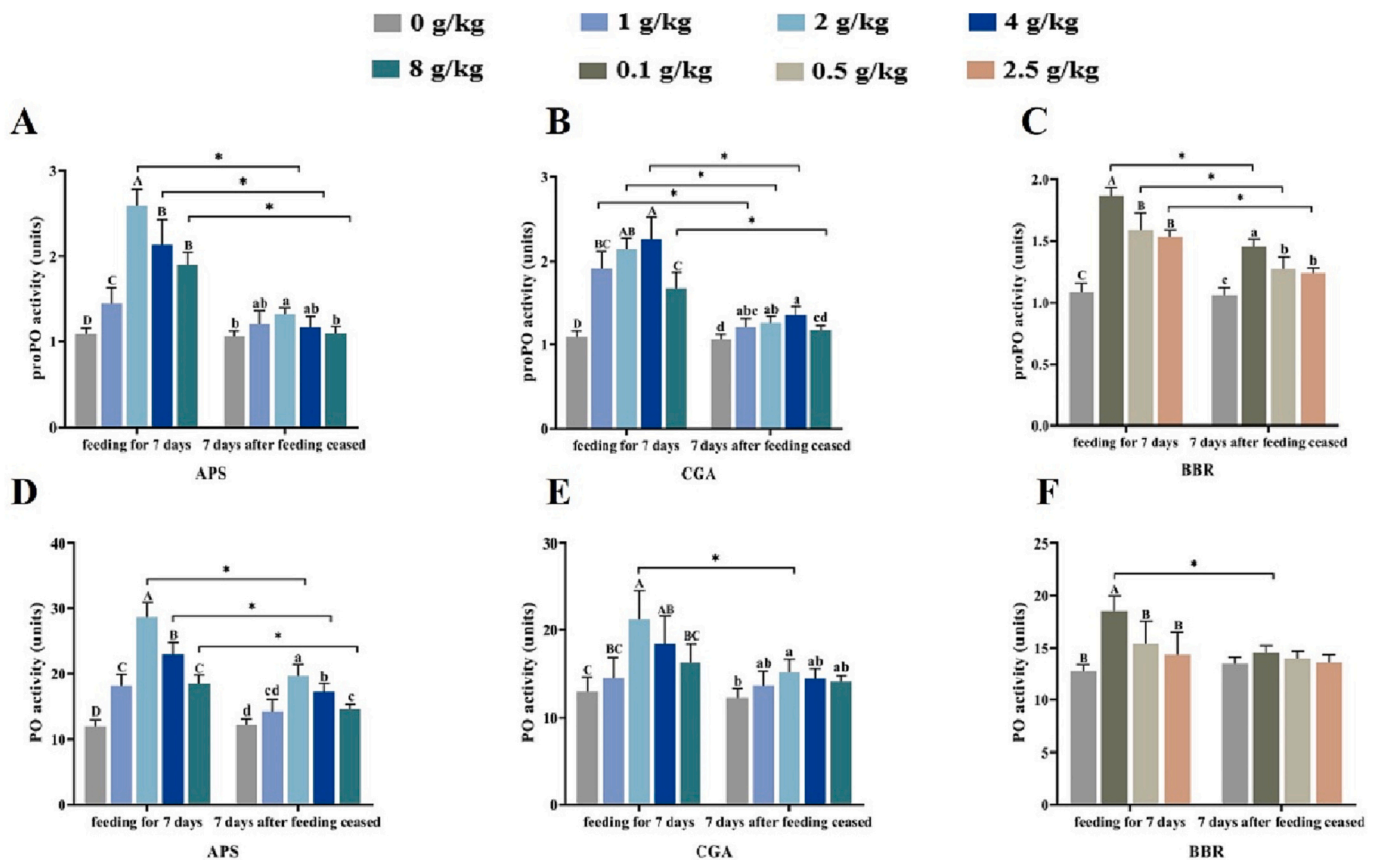


Fig. 2. Effects of *Astragalus* polysaccharides, chlorogenic acid and berberine on the prophenoloxidase activation system of *C. quadricarinatus* after feeding for 7 days and 7 days after feeding was ceased. (A-C: ProPO activity; D-F: PO activity). APS, *Astragalus* polysaccharide; CGA, chlorogenic acid; BBR, berberine; ProPO, prophenoloxidase; PO, phenol oxidase. Data are the mean $\pm$ SD. Bars with different letters represent significant differences ($P < 0.05$). Statistical significance is indicated by * ($P < 0.05$).

3.5. Damage indices

Figure 7 showed the effects of *Astragalus* polysaccharides, chlorogenic acid, and berberine on biological macromolecular damage in crayfish. When *Astragalus* polysaccharides and chlorogenic acid were fed for 7 days, the malondialdehyde content in each treatment group decreased significantly ($P < 0.05$), but the F value and protein carbonyl did not show significant differences compared with those in the control group ($P > 0.05$). Then, 7 days after feeding with *Astragalus* polysaccharides and chlorogenic acid was stopped, the malondialdehyde contents in the experimental groups increased significantly but were still significantly lower than those in the control group ($P < 0.05$). In addition, when berberine was fed for 7 days, the malondialdehyde content in the experimental groups decreased significantly, and when berberine feeding was stopped for 7 days, the malondialdehyde contents increased significantly but were still significantly lower than those in the control group ($P < 0.05$). When berberine was fed for 7 days, the F value did not show a significant difference compared with that in the control group ($P > 0.05$), but the protein carbonyl of berberine 0.5 g/kg and 2.5 g/kg increased significantly ($P < 0.05$).

The effects of *Astragalus* polysaccharides, chlorogenic acid, and berberine on the alanine aminotransferase activity, aspartate aminotransferase activity and hepatopancreas tissue structure of crayfish were shown in Fig. 8. When *Astragalus* polysaccharides and chlorogenic acid were fed for 7 days, the alanine aminotransferase and aspartate aminotransferase activities were not significantly different compared with those in the control group ($P > 0.05$). However, 2.5 g/kg berberine significantly increased alanine aminotransferase and aspartate aminotransferase activities. However, the paraffin sections of the

hepatopancreas in the control group, *Astragalus* polysaccharides 4 g/kg, chlorogenic acid 4 g/kg, and berberine 2.5 g/kg groups were normal, the histological structure of hepatopancreas cells was closely connected, and the central lumen was arranged in a stellate shape.

4. Discussion

The application of Chinese herbal feed additives in the prevention and treatment of aquatic diseases has received extensive attention in recent years. Previous studies have tended to focus on the beneficial effects of Chinese herbal feed additives on the physiological health of organisms. However, there remains a large gap in the study of detoxification and safety evaluation. This study focused on the growth performance, immunity, antioxidant activity, detoxification, and a series of damage indices of crayfish when they were fed with *Astragalus* polysaccharides, chlorogenic acid or berberine for 7 days and not fed these additives for 7 days compared with those of crayfish fed a basic diet to provide reference data for green aquaculture.

4.1. The effects of Chinese herbal feed additives on growth performance

As a feed additive, Chinese herbal medicine can promote the growth of aquatic animals due to its functions in promoting metabolism and the secretion of digestive enzymes (Pu et al., 2017). Research found that the addition of *Radix bupleuri* and *Radix astragali* to the diet could promote the specific growth rate and plumpness of *L. vannamei*. However, the present study showed that after the addition of Chinese herbal medicine, there was no significant difference in growth performance between the control group and the treatment groups. In this study, the reason for this

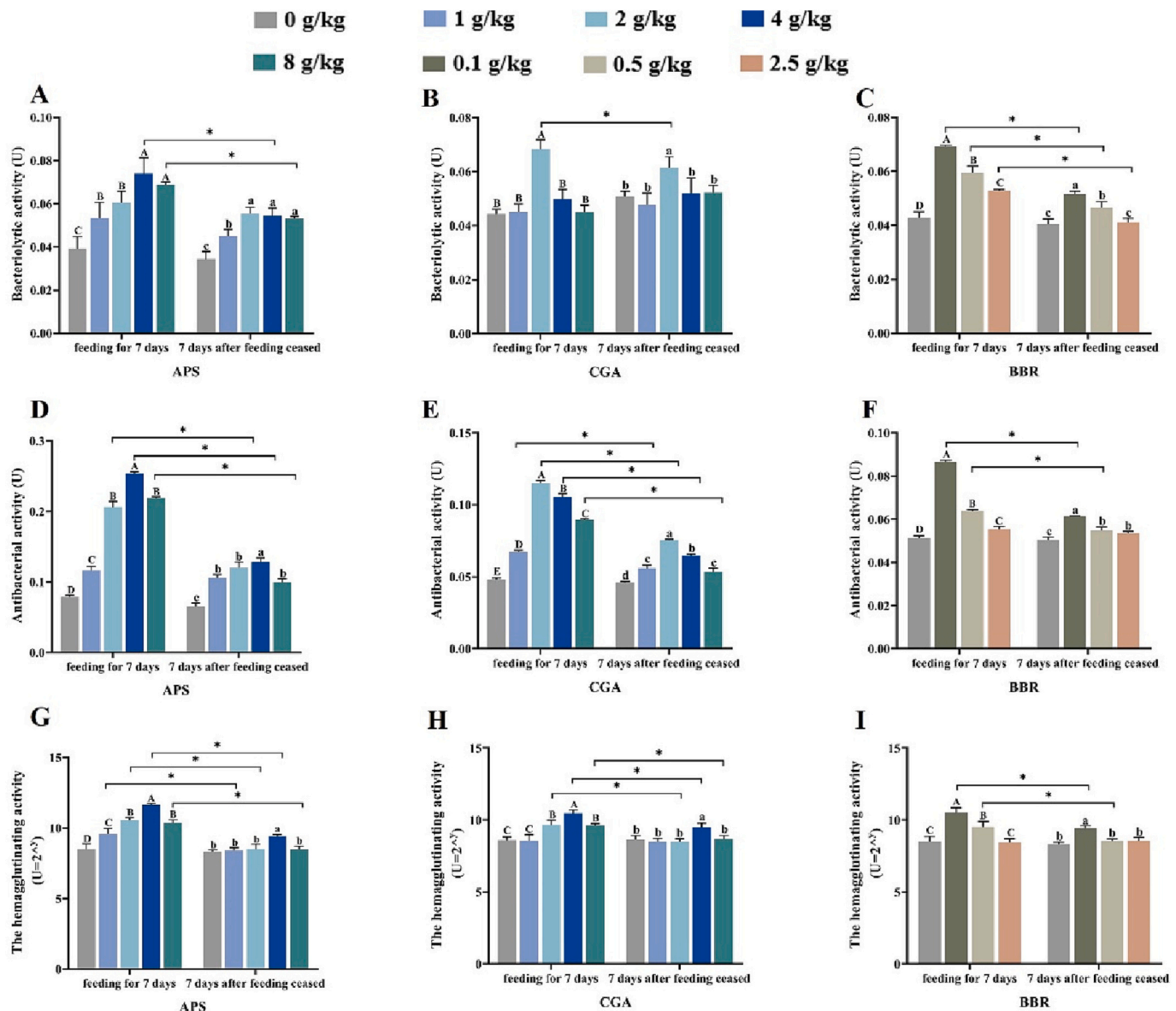


Fig. 3. Effects of bacteriolytic activity, antibacterial activity and agglutination activity of *C. quadricarinatus* after feeding for 7 days and 7 days after feeding was ceased. (A–C: Bacteriolytic activity; D–F: Antibacterial activity; G–I: Agglutination activity). APS, *Astragalus polysaccharide*; CGA, *chlorogenic acid*; BBR, *berberine*. Data are the mean $\pm$ SD. Bars with different letters represent significant differences ($P < 0.05$). Statistical significance is indicated by * ($P < 0.05$).

result may be related to the shorter feeding time of 7 days. Some studies have shown that adding *Sargassum* polysaccharide to the basic diet for 14 days did not have a significant effect on the weight gain and survival of *L. vannamei* (Huang et al., 2006). The body weight and body length of *L. vannamei* were not significantly affected by feeding compound Chinese herbal medicines such as *Astragalus membranaceus* and *Radix isatidis* for 14 days, but there were significant differences in the weight gain rate, specific growth rate, and plumpness of *L. vannamei* after 28 days of feeding (Chen et al., 2017). The growth of *L. vannamei* fed *Astragalus polysaccharide* for 30 days was not obvious (Chang et al., 2018). After examining previous studies, the author concluded that the effects of adding *Astragalus polysaccharides*, *chlorogenic acid*, or *berberine* to crayfish growth performance were not significant due to the short feeding time.

4.2. The effects of Chinese herbal feed additives on immune functions

Crustaceans lack an adaptive immune system and rely only on various innate immune responses to fight against invading pathogens.

Circulating hemocytes, acting as both the undertaker of the cellular immune and the supplier of the humoral immune, play a central role in the immune defense (Zhao et al., 2016). Cellular immunity is an important part of the innate immunity of crustaceans, including phagocytosis, encapsulation, nodule formation, and aggregation (Iwanaga and Lee, 2005; Little and Hultmark, 2005). Phagocytes utilize both oxygen-dependent and oxygen-independent mechanisms to achieve activity against nonself-inducing intracellular particulate particles or microbes. In the oxygen-dependent mechanism, reactive oxygen species (ROS) are generated to damage or kill invading or ingested microorganisms (Hampton et al., 1998; Rosen et al., 1995). However, respiratory burst is a process of rapid oxygen consumption in a short time when phagocytes engulf foreign bodies or are activated by external stimuli (Valavanidis et al., 2005). A study found that *Angelica sinensis* polysaccharide significantly increased the respiratory burst of the hemolymph in *Epinephelus malabaricus* (Wang et al., 2011). Adding berberine to the diet also increased the phagocytosis and respiratory burst activities of the hemolymph in *Oreochromis niloticus* (Doan et al., 2020). The present study found that the hemocyte respiratory burst of crayfish

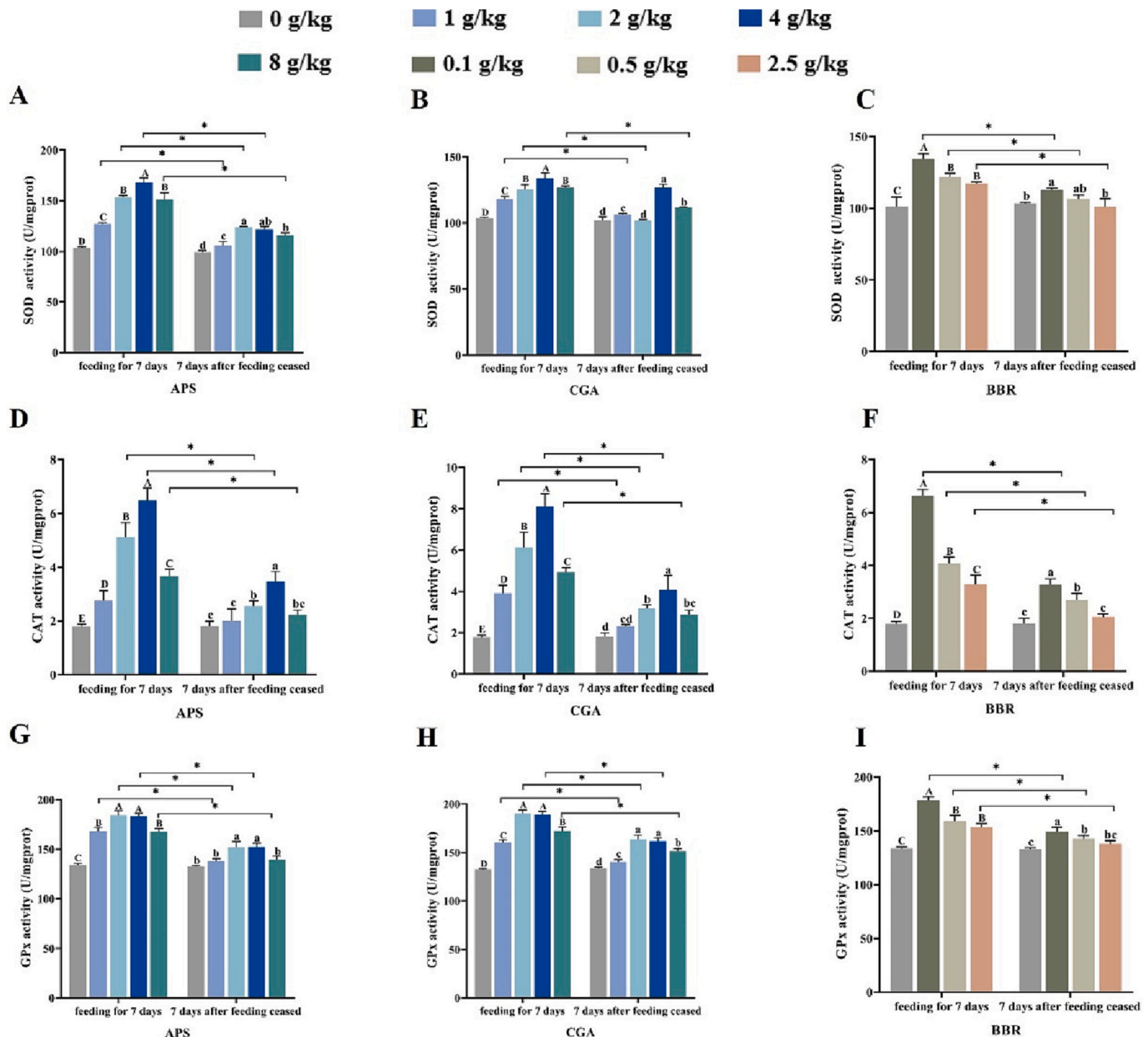


Fig. 4. Effects of *Astragalus* polysaccharides, chlorogenic acid and berberine on antioxidant enzyme activities of *C. quadricarinatus* after feeding for 7 days and 7 days after feeding ceased. (A-C: SOD activity; D-F: CAT activity; G-I: GPx activity). APS, *Astragalus* polysaccharide; CGA, chlorogenic acid; BBR, berberine. Data are the mean $\pm$ SD. Bars with different letters represent significant differences ($P < 0.05$). Statistical significance is indicated by * ($P < 0.05$).

improved significantly and then decreased significantly after stopping feeding bioactive ingredients for 7 days; for the *Astragalus* polysaccharides 2, 4 g/kg, chlorogenic acid 2, 4 g/kg, and berberine 0.1 g/kg groups, the hemocyte respiratory burst was enhanced significantly. In conclusion, the addition of *Astragalus* polysaccharides, chlorogenic acid, or berberine can stimulate the respiratory burst in the process of cell phagocytosis and enhance the cellular immunity of red claw crayfish to some extent by promoting the hemocyte respiratory burst.

Humoral immunity mainly includes the prophenoloxidase activation system and various related immune proteins, such as lysozyme, antimicrobial peptide and lectin (Iwanaga and Lee, 2005; Little and Hultmark, 2005). The prophenoloxidase activation system in crustaceans is an enzymatic reaction system composed of serine protease and prophenoloxidase that exists in hemocytes and can be specifically activated by LPS and β -1,3-glucan (Soderhall and Smith, 1986; Lanz et al., 1993). Many studies have found that adding Chinese herbal medicine to the

basic diet can induce the prophenoloxidase activation system of aquatic animals, thus improving their resistance to disease. The prophenoloxidase activity of *L. vannamei* increased significantly after 7 days of feeding with mixed feed containing multiple Chinese herbal medicines and had a certain protective effect on WSSV infection (El Asely et al., 2010). In this study, the addition of a certain concentration of *Astragalus* polysaccharides, chlorogenic acid, or berberine significantly promoted prophenoloxidase and phenol oxidase activities. This indicates that the addition of *Astragalus* polysaccharides, chlorogenic acid or berberine can promote the prophenoloxidase activation system of crayfish.

Lysozyme is an alkaline protein that can destroy cell walls, release content to dissolve bacteria, and hydrolyze foreign invaders into residues to exclude cells (Stabili et al., 1999). Antimicrobial peptides are short peptides with antibacterial activity and a wide range of universal bactericidal effects that play an important role in resistance to pathogen invasion, making them an important defense component of nonspecific

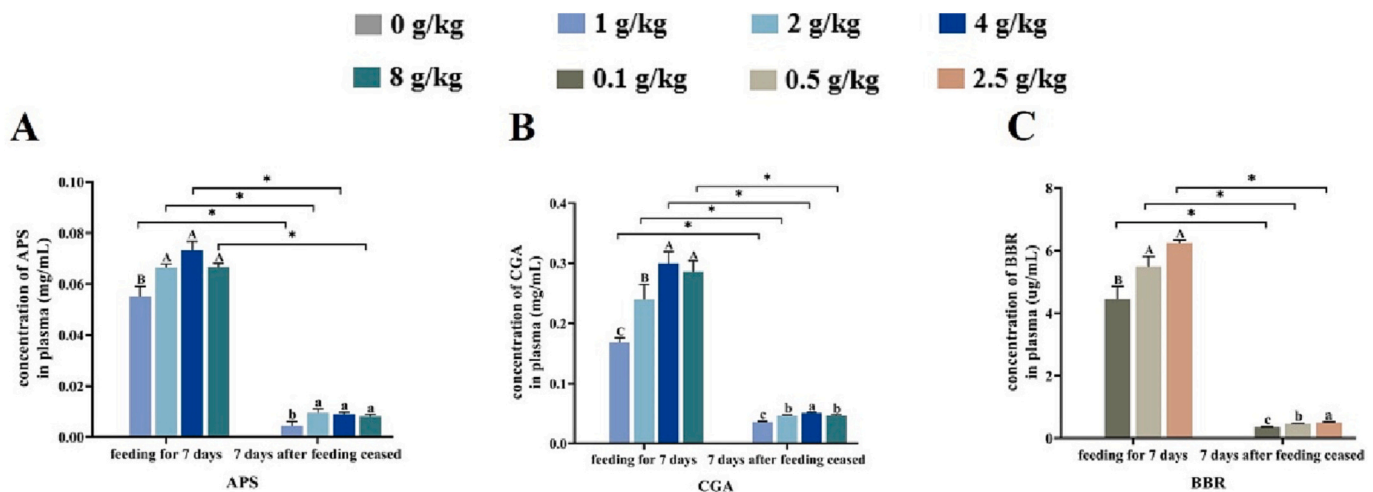


Fig. 5. Effects of *Astragalus* polysaccharides, chlorogenic acid and berberine on the plasma concentration of *C. quadricarinatus* after feeding for 7 days and 7 days after feeding was ceased. (A: APS in the plasma; B: CGA in the plasma; C: BBR in the plasma). APS, *Astragalus* polysaccharide; CGA, chlorogenic acid; BBR, berberine. Data are the mean $\pm$ SD. Bars with different letters represent significant differences ($P < 0.05$). Statistical significance is indicated by * ($P < 0.05$).

immunity (Zhai et al., 2021). Many studies have shown that Chinese herbal medicine additives, such as *Astragalus* polysaccharides, chlorogenic acid, berberine, *Angelica*, *Rehmannia*, *Ligusticum chuanxiong* and *Paeonia lactiflora*, can improve the bacteriolytic activity and antibacterial activity of aquatic animals (Wang et al., 2022; Gao et al., 2017). This study showed that after 7 days of intake of *Astragalus* polysaccharides, chlorogenic acid or berberine, the bacteriolytic activity and antibacterial activity of crayfish were enhanced. Therefore, the author believes that the addition of Chinese herbal medicine can enhance the nonspecific immune defense system of crustaceans to some extent. On the basis of previous studies, the results showed that the addition of Chinese herbal medicine could promote the lysozyme and antimicrobial activity of crayfish.

As an important immune recognition molecule in crustaceans, lectins are essentially a family of proteins with structural heterogeneity and specificity for foreign body binding, which requires Ca^{2+} activation for thermal instability (Hall and Rowlands, 1974a, 1974b). Among them, the C-type lectin family is the largest and most diverse in animals and is also the most widely distributed lectin in crustaceans (Sánchez-Salgado et al., 2017). In *Haliotis discus hamai* Ino, it was found that the ingestion of mixed Chinese herbal preparations such as *tangerine peel* and *hawthorn* significantly increased agglutination activity (Xue et al., 2008). In this study, the intake of a certain dose of *Astragalus* polysaccharides, chlorogenic acid, or berberine can promote the agglutination activity of crayfish, and combined with previous studies on *Apostichopus japonicus* (Wang et al., 2009), it can be concluded that our experimental results are consistent with the current research results. On the basis of previous studies, it can be concluded that adding *Astragalus* polysaccharides, chlorogenic acid, or berberine can promote the agglutination activity of crayfish. In conclusion, *Astragalus* polysaccharides, chlorogenic acid, or berberine could promote the activities of prophenoloxidase, phenol oxidase, bacteriolysis, antibacterial activity and agglutination and comprehensively improve the humoral immunity of red claw crayfish.

4.3. The effects of Chinese herbal feed additives on antioxidant capacity

The hepatopancreas is the most important antioxidant defense organ in crustaceans. When the body is subjected to external stress, it can promote the enzyme-induced and nonenzyme-induced reactions of mitochondria, microsomes and cytoplasm in cells, and the reduction reaction produces reactive oxygen species (Vogt, 2019). The hepatopancreas produces antioxidant enzymes such as superoxide dismutase, catalase and glutathione peroxidase through enzyme-induced reactions

to eliminate excessive reactive oxygen species and help the body reduce the damage caused by oxidative stress (Abele and Puntarulo, 2004; Pani et al., 2000). Superoxide dismutase, as an oxidoreductase, can convert superoxide radicals into H_2O_2 and molecular oxygen to protect cells from the harmful effects of superoxide radicals (Fridovich, 1989). Catalase can effectively utilize hydrogen peroxide and convert it into water and oxygen, which is correlated with superoxide dismutase and can jointly clarify the active oxygen free radicals in organisms (Heck et al., 2010). Glutathione peroxidase protects cells from oxidative damage by reducing lipid peroxides to ethanol and free hydrogen peroxides to water (Ighodaro and Akinloye, 2018; Lubos et al., 2011). Similar results have been reported in *L. vannamei*; under low salt stress, feeding chlorogenic acid can significantly promote total antioxidant capacity, glutathione peroxidase activity, and catalase activity (Wang et al., 2013). Similarly, the antioxidant enzyme activities in fish fed *Astragalus* polysaccharides, chlorogenic acid or berberine increased significantly, such as *Larimichthys crocea* (Zhang et al., 2020), *Danio rerio* (Li et al., 2021), *Trionyx Sinensis* (Zhang, 2021) and *Acanthopagrus schlegelii* (Wang et al., 2020). In this study, it was found that the antioxidant capacity of crayfish reached a maximum in the concentration range of *Astragalus* polysaccharides 2, 4 g/kg, chlorogenic acid 2, 4 g/kg or berberine 0.1 g/kg. The results of this study are consistent with those of previous studies, indicating that the addition of Chinese herb bioactive ingredients can enhance the antioxidant capacity of aquatic organisms and reduce oxidative damage. It is worth noting that chlorogenic acid has a higher antioxidant capacity than *Astragalus* polysaccharides and berberine according to the antioxidant index in this experiment. Huang et al. (2018) confirmed that the antioxidant capacity of the combination of chlorogenic acid and allicin was higher than that of *Astragalus* polysaccharides and allicin, which is consistent with the fact that the antioxidant capacity of chlorogenic acid obtained in this study is higher than that of *Astragalus* polysaccharides, but comparison with berberine has not been fully studied.

4.4. The effects of Chinese herbal feed additives on detoxification

The hepatopancreas also serves as the main metabolic organ of crustaceans (Vogt, 2019). After oral administration, drugs are transported from the stomach to the hepatopancreas, where they are metabolized and eliminated (Tu et al., 2010). The theory that the hepatopancreas can detoxify organic compounds with the help of cytochrome P450 has been repeatedly confirmed in previous studies (Iyanagi, 2007), and glutathione sulfo transferase was also found to be

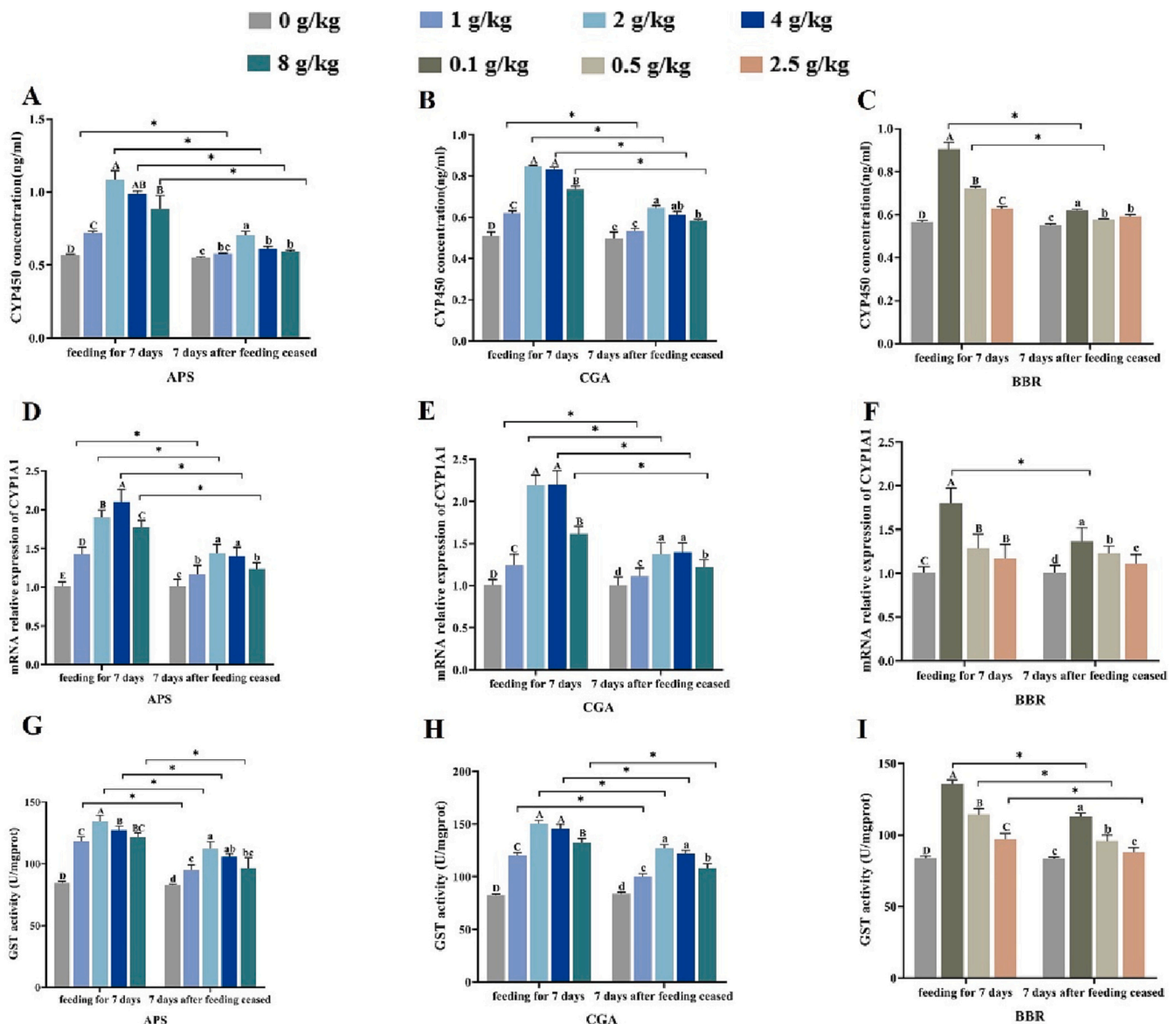


Fig. 6. Effects of *Astragalus* polysaccharides, chlorogenic acid and berberine metabolic enzyme activities of *C. quadricarinatus* after feeding for 7 days and 7 days after feeding was ceased. (A-C: CYP450 concentration; D-F: CYP1A1 mRNA relative expression; G-I: GST activity). APS, *Astragalus* polysaccharide; CGA, chlorogenic acid; BBR, berberine. Data are the mean $\pm$ SD. Bars with different letters represent significant differences ($P < 0.05$). Statistical significance is indicated by * ($P < 0.05$).

involved in detoxification in crustaceans (Nay et al., 1999). Phase I metabolism represented by cytochrome P450 can add active groups to the active ingredients to form intermediate products and then catalyze the intermediate products to form water-soluble metabolites by enhancing the activities of phase II metabolic enzymes, such as glutathione sulfo transferase (Hu et al., 2015; Armstrong, 1997; Ewerc et al., 2019). At present, studies on the metabolism of Chinese herbal medicine in phase I and phase II organisms are mainly focused on mammals, such as rats. The levels of cytochrome P450 and glutathione sulfo transferase activity in the liver of male rats fed with *Ginkgo biloba* extract were found to increase significantly (Ren et al., 2019). In addition, glutathione sulfo transferase activity increased significantly in *L. vannamei* receiving Xiao-Chaihu decoction (Su et al., 2021). Similarly, in this study, it was found that after the intake of *Astragalus* polysaccharides, chlorogenic acid, or berberine, the contents of the Chinese herb bioactive ingredients in vivo increased significantly, and the cytochrome P450 content and glutathione sulfo transferase activity in the hepatopancreas also increased. After seven days, the drug contents in vivo decreased significantly after feeding the control diets, while the cytochrome P450 content and

glutathione sulfo transferase activity in the hepatopancreas also decreased. Therefore, the authors believe that cytochrome P450 and glutathione sulfo transferase are involved in the metabolic process of Chinese herbal medicine in the hepatopancreas, degrading some Chinese herbal medicine in crayfish and reducing the biological damage caused by Chinese herbal medicine to the organisms in the hepatopancreas. Due to their limited degradation capacity, the content of Chinese herbal medicine should be rationally added to the diet of aquatic organisms according to their size, growth cycle, and body weight.

4.5. Safety evaluation of Chinese herbal feed additives

When organisms suffer from environmental stress or pathogen challenge, excessive reactive oxygen species can cause oxidative damage to biological macromolecules such as DNA, proteins, and lipids (Halliwell and Gutteridge, 1999). One type of DNA damage is single-strand breakage, and the DNA alkaline unwinding assay (DAU) is a quick and rapid method for detecting single-strand breakage. The F value is the ratio of the remaining DNA double-strand number to the total DNA after

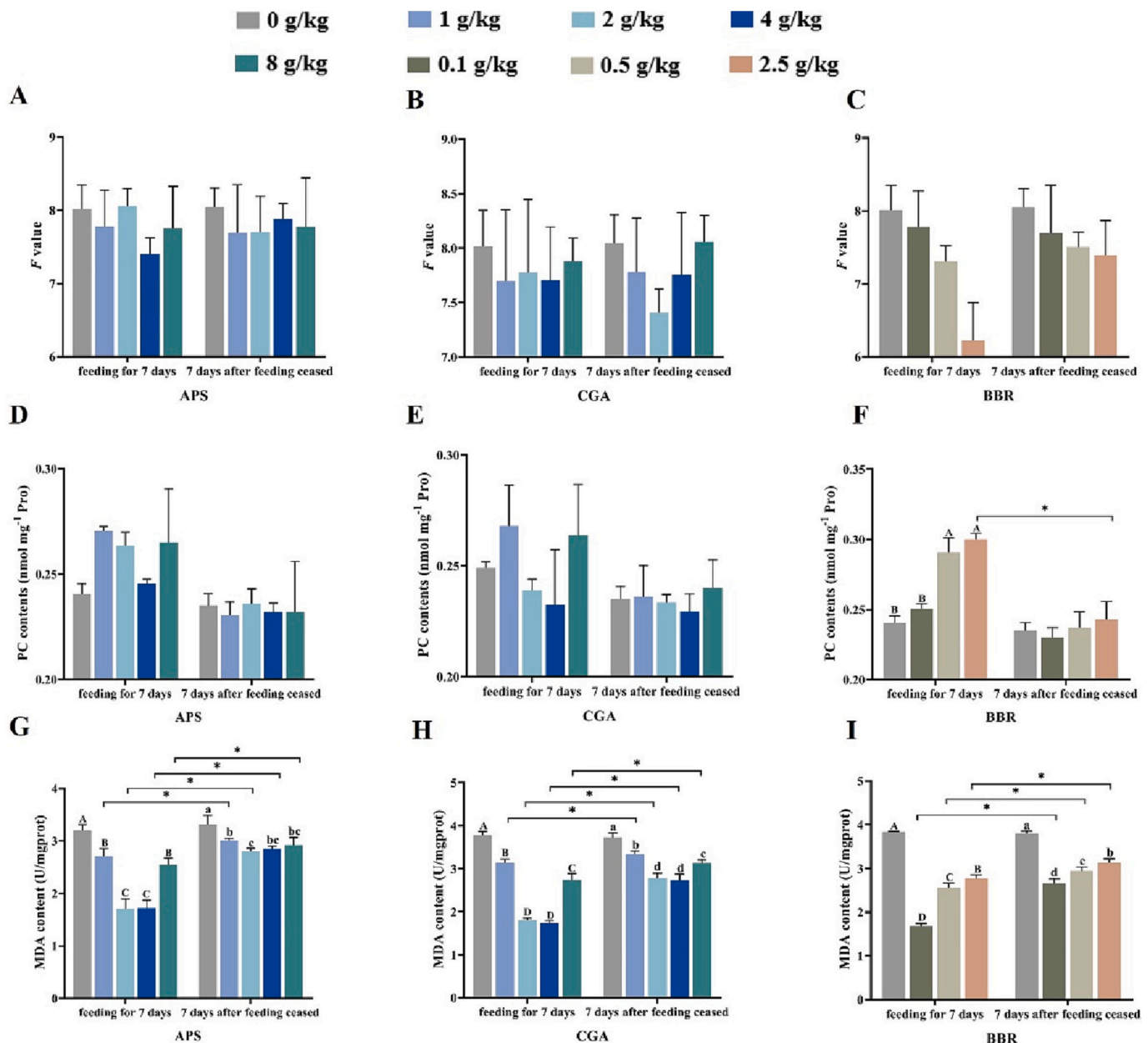


Fig. 7. Effects of *Astragalus* polysaccharides, chlorogenic acid and berberine on the biological macromolecular damage of *C. quadricarinatus* after feeding for 7 days and 7 days after feeding was ceased. (A-C: *F* value; D-F: Protein carbonyl (PC) contents; G-I: MDA content). APS, *Astragalus* polysaccharide; CGA, chlorogenic acid; BBR, berberine. Data are the mean $\pm$ SD. Bars with different letters represent significant differences ($P < 0.05$). Statistical significance is indicated by * ($P < 0.05$).

alkaline hydrolysis. The lower the *F* value is, the higher the possibility of DNA double-strand breaks is (Ding et al., 2020). Likewise, protein carbonylation is an important symbol of free radical oxidation, which is irreversible and can directly lead to permanent loss of protein function (Chora et al., 2008; Dowling et al., 2006). In this study, it was found that feeding with Chinese herbs had no significant differences in DNA damage and protein carbonylation, which was confirmed by previous studies on *L. vannamei* (Huang et al., 2018). Moreover, malondialdehyde is the final product of the lipid peroxidation reaction, and the degree of lipid peroxidation can be evaluated by measuring the malondialdehyde content (Yonar and Sakin, 2011). Many studies have shown that the addition of plant extracts could effectively relieve the level of lipid peroxidation in animals and reduce malondialdehyde content. For example, malondialdehyde content in serum and liver could be reduced to a certain extent in rats that ingest balsam pear polysaccharide (Huang et al., 2020). Moreover, the level of malondialdehyde in the serum of

carp treated with *Rehmannia glutinosa* polysaccharide was significantly lower than that of the control group (Feng et al., 2020). Similarly, in this study, we found that the malondialdehyde content in the hepatopancreas significantly decreased 7 days after ingestion of Chinese herbal medicine, and then the malondialdehyde content gradually increased but remained lower than the control level on Day 7 after stopping feeding with Chinese herb bioactive ingredients. This change is contrary to the trend of antioxidant enzymes mentioned above. This result indicated that the feeding of *Astragalus* polysaccharides, chlorogenic acid, or berberine did not cause oxidative damage to the DNA and protein of macromolecules and inhibited lipid peroxidation to some extent, thereby enhancing the antioxidant capacity of the organism by eliminating excessive free radicals.

Drug-induced liver injury refers to liver injury induced by various chemical drugs, traditional Chinese medicine and its metabolites and even excipients and is one of the most common and serious adverse drug

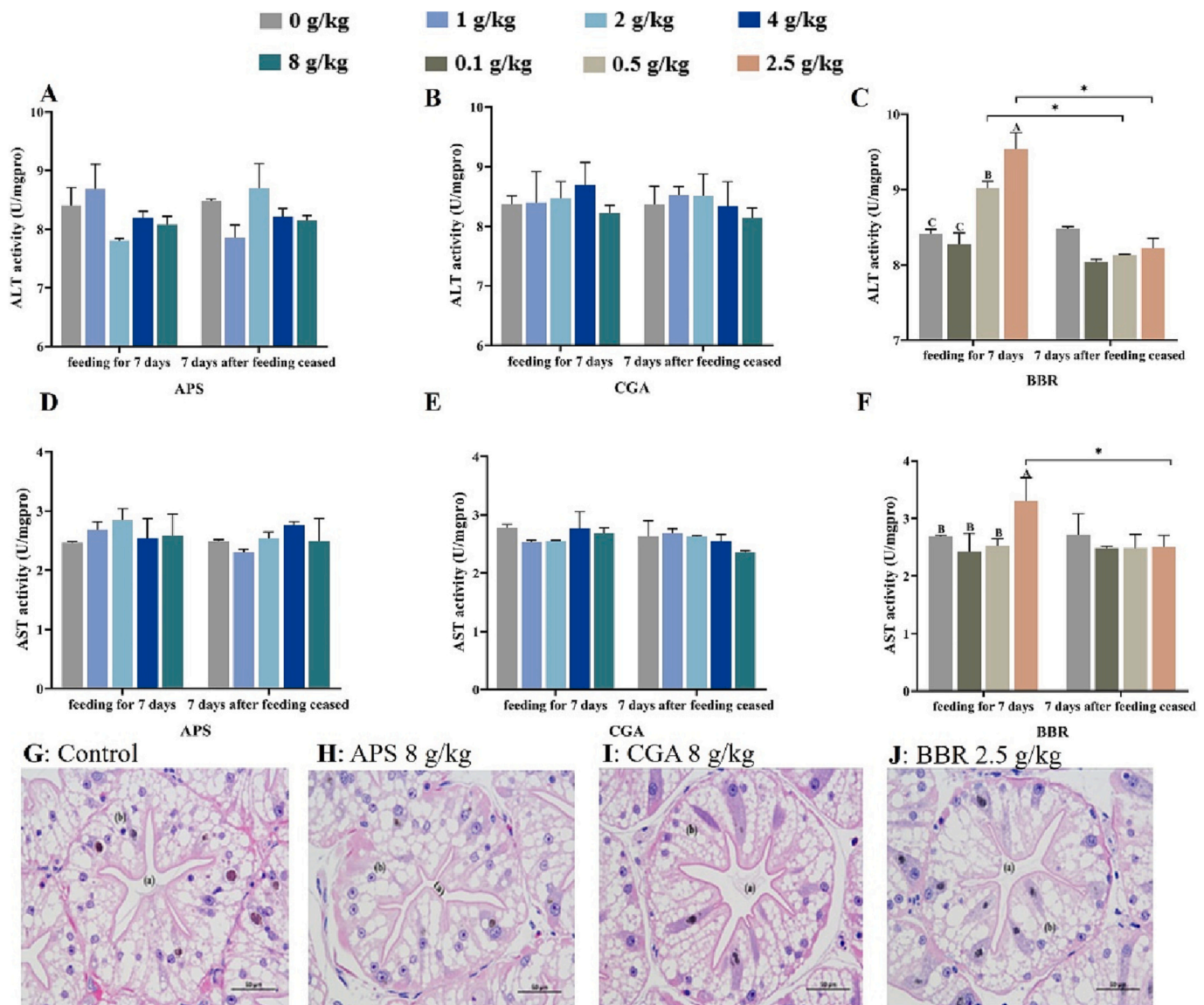


Fig. 8. Effects of *Astragalus* polysaccharides, chlorogenic acid and berberine on alanine aminotransferase activity, aspartate aminotransferase activity and the hepatopancreas tissue structure of *C. quadricarinatus* after feeding for 7 days and 7 days after feeding was ceased. (A-C: ALT activity; D-F: AST activity; G-J: hepatopancreas tissue structure, $\times 400$). The letters in fig. G-J: (a) tubular lumen, (b) B cells. APS, *Astragalus* polysaccharide; CGA, chlorogenic acid; BBR, berberine. Data are the mean $\pm$ SD. Bars with different letters represent significant differences ($P < 0.05$). Statistical significance is indicated by * ($P < 0.05$).

reactions, which is generally seen in the irrational use of drugs (Li et al., 2007). Alanine aminotransferase and aspartate aminotransferase are two important aminotransferases in amino acid metabolism and are widely used to reflect the health levels of cell injury hepatopathy (Batista-Silva et al., 2022). It was found that alanine aminotransferase and aspartate aminotransferase levels were increased significantly after tilapia ingested 5 times the recommended dose of the Chinese herbal medicine “*San Huang Lian*” for 7 days, and pathological damage to the hepatopancreas was obvious, but no death was observed (Huang, 2014). However, feeding within 3 times the dose of *Epimedium brevicornis* Maxim extract for 7 days was safe for crayfish and did not cause pathological damage (Cheng et al., 2020). In the present study, it was found that the addition of *Astragalus* polysaccharides, chlorogenic acid, or berberine 0.1 g/kg did not cause significant changes in alanine aminotransferase and aspartate aminotransferase, but the contents of alanine aminotransferase and aspartate aminotransferase in the hepatopancreas of the berberine 2.5 g/kg group increased significantly. Interestingly, it can be determined from the hepatopancreas tissue sections that the addition of berberine did not cause damage to the hepatopancreas, which is

consistent with previous studies and the results of macromolecular damage mentioned above. Because of the short administration time and low dose of Chinese herb bioactive ingredients in this experiment, they will not cause pathological damage in the hepatopancreas of the crayfish. Through this study, the dose safety of *Astragalus* polysaccharides, chlorogenic acid, and berberine in crayfish was demonstrated from the two aspects of biochemical reactions and apparent effects, and it was found that when *Astragalus* polysaccharides and chlorogenic acid concentrations were < 8 g/kg and that of berberine was < 0.1 g/kg, the crayfish remained healthy and had no pathological damage.

5. Conclusion

The present study showed that Chinese herbal feed additives could significantly improve the immune functions and antioxidant capacity of red claw crayfish, among which *Astragalus* polysaccharides 2, 4 g/kg, chlorogenic acid 2, 4 g/kg, and berberine 0.1 g/kg had the most significant improvement effect. In the detoxification process, the enzyme activities of phase I and II metabolism were significantly enhanced.

Moreover, *Astragalus* polysaccharides, chlorogenic acid, and berberine at low concentrations did not cause damage to crayfish, while berberine at high concentrations (2.5 g/kg) had potential risks, such as protein carbonyl and increased activity of alanine aminotransferase and aspartate aminotransferase enzymes. This study will provide a wealth of knowledge about the physiological effects of *Astragalus* polysaccharides, chlorogenic acid, and berberine in crustaceans and a theoretical basis for the development of immunopotentiator for crayfish.

CRedit authorship contribution statement

Xiaohui Shi: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. **Yongyi Jia:** Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. **Zhuofan Zhang:** Data curation. **Wenbo Wu:** Investigation. **Zijie Wu:** Validation. **Meili Chi:** Data curation. **Qun Zhao:** Writing – review & editing, Project administration. **Erchao Li:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.aquaculture.2023.739394>.

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